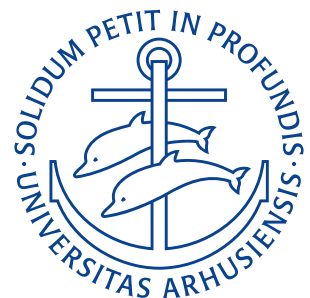


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Homework Exercises

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Contents

1	Lecture 1: Introduction	2
2	Lecture 2: Traction, Stress, and Equilibrium	5
3	Lecture 3: The Cauchy Theorem & Principal Stresses	5
4	Lecture 4: Equilibrium Equations & Principle of Angular Momentum	6
5	Lecture 5: The Strain Tensor	16
6	Lecture 6: Physical interpretation of the strain tensor	18
7	Lecture 7: Volume and shape change	18
8	Lecture 8: Linear isotropic materials and the uniaxial stress	18
9	Lecture 9: Thermal strains	19

Lecture 1: Introduction

26. Januar 2026

Problem 1.1a

Show that

$$\delta_{ij}\delta_{ij} = 3.$$

DerivationExpanding $\delta_{ij}\delta_{ij}$ we get

$$\delta_{ij}\delta_{ij} = \delta_{11}\delta_{11} + \delta_{12}\delta_{12} + \delta_{13}\delta_{13} + \delta_{21}\delta_{21} + \delta_{22}\delta_{22} + \delta_{23}\delta_{23} + \delta_{31}\delta_{31} + \delta_{32}\delta_{32} + \delta_{33}\delta_{33}.$$

As $\delta_{11} = \delta_{22} = \delta_{33} = 1$ the above becomes:

$$\delta_{ij}\delta_{ij} = 1 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 = 3.$$

$\delta_{ij}\delta_{ij} = 3$

RESULT

Problem 1.1c

Show that

$$\alpha_{ij}b_{jk} = \alpha_{in}b_{nk}.$$

Derivation

We expand by summing over the common index j :

$$\alpha_{ij}b_{jk} = \sum_j \alpha_{ij}b_{jk} = \alpha_{i1}b_{1k} + \alpha_{i2}b_{2k} + \alpha_{i3}b_{3k}.$$

Now we sum over n instead:

$$\alpha_{i1}b_{1k} + \alpha_{i2}b_{2k} + \alpha_{i3}b_{3k} = \sum_n \alpha_{in}b_{nk}.$$

Q.E.D.

$\alpha_{ij}b_{jk} = \alpha_{in}b_{nk}$

RESULT

Problem 1.1e

Show that

$$\delta_{ij}C_{jkl} = C_{ikl}.$$

Derivation

Expanding we get

$$\delta_{ij}C_{jkl} = \delta_{11}C_{1kl} + \delta_{22}C_{2kl} + \delta_{33}C_{3kl} = C_{ikl}.$$

$\delta_{ij}C_{jkl} = C_{ikl}$

RESULT

Lecture 2: Traction, Stress, and Equilibrium

2. Februar 2026

Lecture 3: The Cauchy Theorem & Principal Stresses

9. Februar 2026

Problem 2.1

Find the components of the traction on a plane defined by $n_1 = 1/\sqrt{2}$, $n_2 = 1/\sqrt{2}$, and $n_3 = 0$ for the following states of stress:

- (a) $\sigma_{11} = \sigma, \sigma_{12} = 0, \sigma_{13} = 0, \sigma_{21} = \sigma, \sigma_{22} = 0, \sigma_{33} = \sigma$.
 (b) $\sigma_{11} = \sigma, \sigma_{12} = \sigma, \sigma_{13} = 0, \sigma_{22} = \sigma, \sigma_{23} = 0, \sigma_{33} = 0$.

Derivation

- (a) The stress state in (a) is

$$\sigma = \begin{bmatrix} \sigma & 0 & 0 \\ \sigma & 0 & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma \end{bmatrix}.$$

The Cauchy Theorem states that

$$T_i = \sigma_{ji} n_j \implies T = \sigma \cdot n.$$

Our plane is:

$$n = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}.$$

Hence,

$$T = \begin{bmatrix} \sigma & 0 & 0 \\ \sigma & 0 & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} = \begin{pmatrix} \frac{\sigma}{\sqrt{2}} \\ \frac{\sigma}{\sqrt{2}} \\ \frac{\sigma_{31}}{\sqrt{2}} + \frac{\sigma_{32}}{\sqrt{2}} \end{pmatrix}.$$

- (b) For (b) our stress state is:

$$\sigma = \begin{bmatrix} \sigma & \sigma & 0 \\ \sigma_{21} & \sigma & 0 \\ \sigma_{31} & \sigma_{32} & 0 \end{bmatrix},$$

so

$$T = \begin{bmatrix} \sigma & \sigma & 0 \\ \sigma_{21} & \sigma & 0 \\ \sigma_{31} & \sigma_{32} & 0 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} = \begin{pmatrix} \sqrt{2}\sigma \\ \frac{\sigma}{\sqrt{2}} + \frac{\sigma_{21}}{\sqrt{2}} \\ \frac{\sigma_{31}}{\sqrt{2}} + \frac{\sigma_{32}}{\sqrt{2}} \end{pmatrix}.$$

(a) $T = \begin{bmatrix} \frac{\sigma}{\sqrt{2}} \\ \frac{\sigma}{\sqrt{2}} \\ \frac{\sigma_{31}}{\sqrt{2}} + \frac{\sigma_{32}}{\sqrt{2}} \end{bmatrix}$

(b) $T = \begin{bmatrix} \sqrt{2}\sigma \\ \frac{\sigma}{\sqrt{2}} + \frac{\sigma_{21}}{\sqrt{2}} \\ \frac{\sigma_{31}}{\sqrt{2}} + \frac{\sigma_{32}}{\sqrt{2}} \end{bmatrix}.$

RESULT

Problem 2.6

At a given point in a body,

$$\sigma_{11} = \sigma_{22} = \sigma_{33} = -p,$$

$$\sigma_{12} = \sigma_{23} = \sigma_{31} = 0.$$

Show that the normal stresses are equal to $-p$ and that the shearing stresses vanish for any other Cartesian coordinate system.

Derivation

Here, our stress tensor is

$$\boldsymbol{\sigma} = \begin{bmatrix} -p & 0 & 0 \\ 0 & -p & 0 \\ 0 & 0 & -p \end{bmatrix} = -p\mathbf{I}.$$

Using the Cauchy theorem $\mathbf{T} = \boldsymbol{\sigma} \cdot \mathbf{n}$ we get:

$$\mathbf{T} = (-p\mathbf{I}) \cdot \mathbf{n} = -p\mathbf{n}.$$

Hence, $\mathbf{T}$ is parallel to $\mathbf{n}$ meaning the traction does not have any components tangent to the plane and hence the stress is entirely normal with size $-p$.

Lecture 4: Equilibrium Equations & Principle of Angular Momentum

16. Februar 2026

Problem 2.2

The state of stress at a point P in a structure is given by

$$\sigma_{11} = 20000,$$

$$\sigma_{22} = -15000,$$

$$\sigma_{33} = 3000$$

$$\sigma_{12} = 2000,$$

$$\sigma_{23} = 2000,$$

$$\sigma_{31} = 1000.$$

- Compute the scalar components T_1 , T_2 , and T_3 of the traction $\mathbf{T}$ on the plane passing through P whose outward normal vector $\mathbf{n}$ makes equal angles with the coordinate axes x_1 , x_2 , x_3 .
- Compute the normal and tangential components of stress on this plane.
- Determine the principal stresses.
- Determine the direction of maximum principal stress.

Derivation

- We are given the following stress state:

$$\boldsymbol{\sigma} = \begin{bmatrix} 20000 & 2000 & \cdot \\ \cdot & -15000 & 2000 \\ 1000 & \cdot & 3000 \end{bmatrix}.$$

We assume this is symmetric i.e. it satisfies the standard case of $\sigma_{ij} = \sigma_{ji}$ and we therefore get:

$$\boldsymbol{\sigma} = \begin{bmatrix} 20000 & 2000 & 1000 \\ 2000 & -15000 & 2000 \\ 1000 & 2000 & 3000 \end{bmatrix}.$$

As $\mathbf{n}$ makes equal angles with x_1 , x_2 , and x_3 , $\mathbf{n}$ must be:

$$\mathbf{n} = \frac{1}{\sqrt{3}}(1, 1, 1).$$

The traction vector on the plane is:

$$\mathbf{T} = \boldsymbol{\sigma} \mathbf{n}$$

so,

$$\begin{aligned} T_1 &= \frac{1}{\sqrt{3}} (20000 + 2000 + 1000) = \frac{23000}{\sqrt{3}} \\ T_2 &= \frac{1}{\sqrt{3}} (2000 - 15000 + 2000) = -\frac{11000}{\sqrt{3}} \\ T_3 &= \frac{1}{\sqrt{3}} (1000 + 2000 + 3000) = \frac{6000}{\sqrt{3}}. \end{aligned}$$

(b) The normal stress is given as

$$\sigma_{nn} = \mathbf{T}_n \cdot \mathbf{n}.$$

I.e.

$$\begin{aligned} \sigma_{nn} &= \frac{1}{\sqrt{3}} (T_1 + T_2 + T_3) \\ &= \frac{1}{3} (23000 - 11000 + 6000) \\ &= 6000. \end{aligned}$$

Hence, the normal traction vector $\mathbf{N}$ is

$$\mathbf{N} = \sigma_{nn} \cdot \mathbf{n} = 6000 \cdot \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

As the total traction $\mathbf{T}$ is the algebraic sum of the normal traction $\mathbf{N}$ and the tangential traction $\mathbf{S}$ we get:

$$\mathbf{T} = \mathbf{N} + \mathbf{S} \implies \mathbf{S} = \mathbf{T} - \mathbf{N}.$$

So

$$\mathbf{S} = \frac{1}{\sqrt{3}} \begin{pmatrix} 23000 \\ -11000 \\ 6000 \end{pmatrix} - \frac{6000}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{3}} \begin{pmatrix} 17000 \\ -17000 \\ 0 \end{pmatrix}.$$

(d) For principal stresses we have the condition:

$$|\sigma_{ji} - \lambda \delta_{ij}| = 0$$

i.e.:

$$\begin{vmatrix} 20000 - \lambda & 2000 & 1000 \\ 2000 & -15000 - \lambda & 2000 \\ 1000 & 2000 & 3000 - \lambda \end{vmatrix} = 0 \rightarrow \lambda_1 \rightarrow -15319,7, \quad \lambda_2 \rightarrow 3133,4, \quad \lambda_3 \rightarrow 20186,3.$$

Therefore our principal stresses are:

$$\sigma^{(1)} = -15319,7$$

$$\sigma^{(2)} = 3133,4$$

$$\sigma^{(3)} = 20186,3.$$

(e) For the maximum principal stress $\lambda_3 = 20186,3$ we solve

$$(\sigma_{ji} - \lambda \delta_{ij}) n_j = 0.$$

I.e.

$$\begin{pmatrix} 20000 - 20186,3 & 2000 & 1000 \\ 2000 & -15000 - 20186,3 & 2000 \\ 1000 & 2000 & 3000 - 20186,3 \end{pmatrix} \cdot \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Solving this gives $n_1 = 15,3299$, $n_2 = 0,928198$, $n_3 = 1$. Now we scale this by the length to get a unit vector:

$$\mathbf{n}_{\max} = \frac{\begin{pmatrix} 15,3299 \\ 0,928198 \\ 1 \end{pmatrix}}{\sqrt{15,3299^2 + 0,928198^2 + 1^2}} = \begin{pmatrix} 0,996 \\ 0,0603 \\ 0,0650 \end{pmatrix}.$$

(a) $\mathbf{T} = \frac{1}{\sqrt{3}} \begin{pmatrix} 23000 \\ -11000 \\ 6000 \end{pmatrix}$

(b) $\mathbf{N} = \frac{1}{\sqrt{3}} \begin{pmatrix} 6000 \\ 6000 \\ 6000 \end{pmatrix}$

$\mathbf{S} = \frac{1}{\sqrt{3}} \begin{pmatrix} 17000 \\ -17000 \\ 0 \end{pmatrix}$

(d) $\sigma^{(1)} = -15319,7$
 $\sigma^{(2)} = 3133,4$
 $\sigma^{(3)} = 20186,3$

(e) $\mathbf{n}_{\max} = \begin{pmatrix} 0,996 \\ 0,060 \\ 0,065 \end{pmatrix}.$

RESULT

Problem 2.3

The state of stress at a point is given by the components of stress tensor σ_{ij} . A plane is defined by the direction cosines of the normal $(1/2, 1/2, 1/\sqrt{2})$. State the general conditions for which the traction on the plane has the same direction as the x_2 -axis and a magnitude of 1,0.

Derivation

We use

$$T_i = \sigma_{ji} n_j = \boldsymbol{\sigma} \cdot \mathbf{n}.$$

We are given

$$\mathbf{n} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{\sqrt{2}} \end{pmatrix}.$$

The traction should be in the x_2 axis and have magnitude 1,0 so it must be

$$\mathbf{T} = \begin{pmatrix} 0 \\ 1,0 \\ 0 \end{pmatrix} = \mathbf{e}_2.$$

So the condition is simply

$$\boldsymbol{\sigma} \mathbf{n} = \mathbf{e}_2.$$

$\boldsymbol{\sigma} \mathbf{n} = \mathbf{e}_2$

RESULT

Problem 2.4

Determine the body forces for which the following stress field describes a state of equilibrium

$$\sigma_{xx} = -2x^2 - 3y^2 - 5z,$$

$$\sigma_{xy} = x + 4xy - 6,$$

$$\sigma_{yy} = -2y^2 + 7,$$

$$\sigma_{xz} = -3x + 2y + 1.$$

$$\sigma_{zz} = 4x + y + 3z - 5,$$

$$\sigma_{yz} = 0.$$

Derivation

We have that

$$\sigma_{ij,j} + f_i = 0.$$

which is:

$$\begin{aligned} \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z} + f_x &= 0 \\ \frac{\partial \sigma_{yx}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{yz}}{\partial z} + f_y &= 0 \\ \frac{\partial \sigma_{zx}}{\partial x} + \frac{\partial \sigma_{zy}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} + f_z &= 0. \end{aligned}$$

In the x -direction we get

$$\begin{aligned} \frac{\partial(-2x^2 - 3y^2 - 5z)}{\partial x} + \frac{\partial(x + 4xy - 6)}{\partial y} + \frac{\partial(-3x + 2y + 1)}{\partial z} + f_x &= 0 \\ -4x + 4x + f_x &= 0 \\ f_x &= 0. \end{aligned}$$

In the y -direction we get

$$\begin{aligned} \frac{\partial(x + 4xy - 6)}{\partial x} + \frac{\partial(-2y^2 + 7)}{\partial y} + \frac{\partial 0}{\partial z} + f_y &= 0 \\ 1 + 4y - 4y + f_y &= 0 \\ f_y &= -1. \end{aligned}$$

And in the z -direction we get

$$\begin{aligned} \frac{\partial(-3x + 2y + 1)}{\partial x} + \frac{\partial 0}{\partial y} + \frac{\partial(4x + y + 3z - 5)}{\partial z} + f_z &= 0 \\ -3 + 3 + f_z &= 0 \\ f_z &= 0. \end{aligned}$$

$$\mathbf{f} = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} = -\mathbf{e}_2$$

RESULT

Problem 2.5

Determine whether the following stress field is admissible in an elastic body when body forces are negligible.

$$\boldsymbol{\sigma} = \begin{bmatrix} yz+4 & z^2+2x & 5y+z \\ \cdot & xz+3y & 8x^3 \\ \cdot & \cdot & 2xyz \end{bmatrix}.$$

Derivation

We once again use

$$\sigma_{ij,j} + f_i = 0.$$

With negligible body forces this becomes

$$\sigma_{ij,j} = 0.$$

I.e.

$$\begin{aligned} \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z} &= 0 \\ \frac{\partial \sigma_{yx}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{yz}}{\partial z} &= 0 \\ \frac{\partial \sigma_{zx}}{\partial x} + \frac{\partial \sigma_{zy}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} &= 0. \end{aligned}$$

In the x -direction we get

$$\begin{aligned} \frac{\partial(yz+4)}{\partial x} + \frac{\partial(z^2+2x)}{\partial y} + \frac{\partial(5y+z)}{\partial z} &= 0 \\ 0 + 0 + 1 &\neq 0. \end{aligned}$$

Hence, the stress field is not admissible with negligible body forces.

It is not.

RESULT

Problem 2.8

The stress tensor at a point $P(x, y, z)$ in a solid is

$$\boldsymbol{\sigma} = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}.$$

- Show that the principal stresses are 4, 1, and -2 .
- Find the orientation of the principal planes

Derivation

- We have the characteristic equation

$$|\sigma_{ji} - \lambda \delta_{ij}| = 0.$$

I.e.

$$\begin{vmatrix} 3-\lambda & 1 & 1 \\ 1 & 0-\lambda & 2 \\ 1 & 2 & 0-\lambda \end{vmatrix} = 0 \Rightarrow \lambda_1 \rightarrow -2, \quad \lambda_2 \rightarrow 1, \quad \lambda_3 \rightarrow 4.$$

- The orientation of the principal planes are the eigenvectors of these. I.e. we have to solve

$$(\sigma_{ji} - \lambda \delta_{ij}) n_j = 0.$$

For each λ .

For $\lambda_1 = -2$ we get

$$\begin{bmatrix} 5 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 2 \end{bmatrix} \cdot \mathbf{n}_1 = 0 \Rightarrow \mathbf{n}_1 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}.$$

$$\hat{\mathbf{n}}_1 = \begin{pmatrix} 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}.$$

For $\lambda_2 = 1$ we get

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & -1 & 2 \\ 1 & 2 & -1 \end{bmatrix} \mathbf{n}_2 = 0 \Rightarrow \mathbf{n}_2 = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}.$$

$$\hat{\mathbf{n}}_2 = \begin{pmatrix} -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}.$$

For $\lambda_3 = 4$ we get

$$\begin{bmatrix} -1 & 1 & 1 \\ 1 & -4 & 2 \\ 1 & 2 & -4 \end{bmatrix} \mathbf{n}_3 = 0 \Rightarrow \mathbf{n}_3 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}.$$

$$\hat{\mathbf{n}}_3 = \begin{pmatrix} \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{pmatrix}.$$

$$(a) \quad \sigma^{(1)} = -2$$

$$\sigma^{(2)} = 1$$

$$\sigma^{(3)} = 4$$

$$(b) \quad \hat{n}_1 = \begin{pmatrix} 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$\hat{n}_2 = \begin{pmatrix} -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$\hat{n}_3 = \begin{pmatrix} \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{pmatrix}.$$

RESULT

Problem 2.9

The stress tensor at a point $P(x, y, z)$ in a solid is

$$\boldsymbol{\sigma} = \begin{bmatrix} 120 & 40 & 30 \\ 40 & 150 & 20 \\ 30 & 20 & 100 \end{bmatrix}.$$

- (a) Determine the stresses with respect to a set of axes x' , y' , and z' for which the matrix of direction cosines is

$$[\alpha_{ij}] = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- (b) Compute the principal stresses using both the x -, y -, and z - and x' -, y' -, and z' -axes as the basis.
 (c) Find the orientation of the principal planes with respect to the x -, y -, and z -axes.

Derivation

- (a) To transform coordinates we use

$$\mathbf{A}' = \mathbf{R} \mathbf{A} \mathbf{A}^T.$$

Hence

$$\boldsymbol{\sigma}' = \boldsymbol{\alpha} \boldsymbol{\sigma} \boldsymbol{\alpha}^T$$

which gives

$$\boldsymbol{\sigma}' = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 120 & 40 & 30 \\ 40 & 150 & 20 \\ 30 & 20 & 100 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{bmatrix} 107,86 & -32,99 & -2,32 \\ -32,99 & 162,14 & 35,98 \\ -2,32 & 35,98 & 100 \end{bmatrix}.$$

- (b) Here we use the characteristic equation from before to find the eigenvalues:

$$\begin{bmatrix} 120 - \lambda & 40 & 30 \\ 40 & 150 - \lambda & 20 \\ 30 & 20 & 100 - \lambda \end{bmatrix} = 0 \implies \lambda_1 = 190,405, \quad \lambda_2 = 102,423, \quad \lambda_3 = 77,1724.$$

and

$$\begin{bmatrix} 107,86 - \lambda & -32,99 & -2,32 \\ -32,99 & 162,14 - \lambda & 35,98 \\ -2,32 & 35,98 & 100 - \lambda \end{bmatrix} \rightarrow \lambda_1 = 190,403, \quad \lambda_2 = 102,424, \quad \lambda_3 = 77,1727.$$

Which confirms that principal stresses are unchanged by rotation.

- (c) Here we solve

$$|\sigma_{ji} - \lambda \delta_{ij}| n_j = 0$$

for each eigenvalue and get:

$$\begin{bmatrix} 120 & 40 & 30 \\ 40 & 150 & 20 \\ 30 & 20 & 100 \end{bmatrix} \rightarrow \left(\begin{array}{ccc} 190.405 & 102.423 & 77.1724 \\ \{1.61658, 2.09537, 1.\} & \{0.821389, -1.11094, 1.\} & \{-0.911651, 0.226096, 1.\} \end{array} \right).$$

Now we scale each $\mathbf{n}$ and get:

$$\mathbf{n}_1 = \begin{pmatrix} 0,5714 \\ 0,7406 \\ 0,3535 \end{pmatrix}$$

$$\mathbf{n}_2 = \begin{pmatrix} 0,4816 \\ -0,6514 \\ 0,5863 \end{pmatrix}$$

$$\mathbf{n}_3 = \begin{pmatrix} 0,6645 \\ -0,1648 \\ -0,7289 \end{pmatrix}.$$

$$(a) \quad \boldsymbol{\sigma} = \begin{pmatrix} 107,86 & -32,99 & -2,32 \\ -32,99 & 162,14 & 35,98 \\ -2,32 & 35,98 & 100 \end{pmatrix}$$

$$(b) \quad \lambda_1 = 190,405$$

$$\lambda_2 = 102,423$$

$$\lambda_3 = 77,1724$$

$$(c) \quad \mathbf{n}_1 = \begin{pmatrix} 0,5714 & 0,7406 & 0,3535 \end{pmatrix}$$

$$\mathbf{n}_2 = \begin{pmatrix} 0,4816 & -0,6514 & 0,5863 \end{pmatrix}$$

$$\mathbf{n}_3 = \begin{pmatrix} 0,6645 & -0,1648 & -0,7289 \end{pmatrix}.$$

RESULT

Lecture 5: The Strain Tensor

23. Februar 2026

Problem 3.1

The components of a displacement field are

$$\begin{aligned}u_x &= (x^2 + 20) \cdot 10^{-4}, \\u_y &= 2yz \cdot 10^{-3}, \\u_z &= (z^2 - xy) \cdot 10^{-3}.\end{aligned}$$

- (a) Consider two points in the undeformed system, $(2, 5, 7)$ and $(3, 8, 9)$. Find the change in distance between these points.

Derivation

The displacement at $A = (2, 5, 7)$ is

$$\begin{aligned}u_x(A) &= (2^2 + 20) \cdot 10^{-4} = 0,0024 \\u_y(A) &= 2 \cdot 5 \cdot 7 \cdot 10^{-3} = 0,070 \\u_z(A) &= (7^2 - 2 \cdot 5) \cdot 10^{-3} = 0,039\end{aligned}$$

so

$$\mathbf{u}(A) = \begin{pmatrix} 0,0024 \\ 0,070 \\ 0,039 \end{pmatrix}.$$

And at $B = (3, 8, 9)$ it is

$$\begin{aligned}u_x(B) &= (3^2 + 20) \cdot 10^{-4} = 0,0029 \\u_y(B) &= 2 \cdot 8 \cdot 9 \cdot 10^{-3} = 0,144 \\u_z(B) &= (9^2 - 3 \cdot 8) \cdot 10^{-3} = 0,057\end{aligned}$$

so

$$\mathbf{u}(B) = \begin{pmatrix} 0,0029 \\ 0,144 \\ 0,057 \end{pmatrix}.$$

The deformed positions are:

$$\begin{aligned}A' &= A + \mathbf{u}(A) = \begin{pmatrix} 2 \\ 5 \\ 7 \end{pmatrix} + \begin{pmatrix} 0,0024 \\ 0,070 \\ 0,039 \end{pmatrix} = \begin{pmatrix} 2,0024 \\ 5,070 \\ 7,039 \end{pmatrix} \\B' &= B + \mathbf{u}(B) = \begin{pmatrix} 3 \\ 8 \\ 9 \end{pmatrix} + \begin{pmatrix} 0,0029 \\ 0,144 \\ 0,057 \end{pmatrix} = \begin{pmatrix} 3,0029 \\ 8,144 \\ 9,057 \end{pmatrix}.\end{aligned}$$

Our position vectors are:

$$AB = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$$

$$AB' = \begin{pmatrix} 1.0005 \\ 3.074 \\ 2.018 \end{pmatrix}.$$

These have the lengths:

$$|AB| = \sqrt{1^2 + 3^2 + 2^2} = 3.74166$$

$$|AB'| = \sqrt{1.0005^2 + 3.074^2 + 2.018^2} = 3.81088.$$

The change in length is then:

$$\Delta L = |AB'| - |AB| = 3.81088 - 3.74166 = 0.06922.$$

$\Delta L = 0.06922$

RESULT

Lecture 6: Physical interpretation of the strain tensor

2. Marts 2026

Problem 3.2

The strain field at a point $P(x, y, z)$ in an elastic body is given by

$$\epsilon = \begin{bmatrix} 20 & 3 & 2 \\ 3 & -10 & 5 \\ 2 & 5 & -8 \end{bmatrix} \times 10^{-6}.$$

Determine the following values:

- (a) The strain invariants
- (b) The principal strains
- (e) The mean normal strain and the strain deviator

Problem 3.3

With reference to Problem 3.2, consider a second coordinate system x'_i in which x'_2 is parallel to x_2 and x'_1 and x'_3 are defined by a 60° counterclockwise rotation about x_2 . Find the components of strain with reference to the x'_i system using the strain field from Problem 3.2.

Problem 3.6

Compute the infinitesimal strains, and sketch the deformed configuration of an initially rectangular volume element subject to the following displacement fields, where A , B , and D_{ij} = constants:

- (a) Simple extension: $u_1 = A_1 x_1; u_2 = u_3 = 0$.
- (b) Simple shear: $u_1 = B x_2; u_2 = u_3 = 0$.
- (c) homogeneous deformation: $u_i = D_{ij} x_j$.

Lecture 7: Volume and shape change

9. Marts 2026

Lecture 8: Linear isotropic materials and the uniaxial stress

16. Marts 2026

Problem 4.2

For an isotropic material, obtain the relationship

$$\epsilon_{ij} = \frac{1}{2\mu} \sigma_{ij} - \frac{\lambda}{2\mu(2\mu + 3\lambda)} \delta_{ij} \sigma_{kk}$$

from the expression

$$\sigma_{ij} = 2\mu \epsilon_{ij} + \lambda \delta_{ij} \epsilon_{kk}.$$

Problem 4.3a

Show that the principal axes of stress correspond with the principal axes of strain for an isotropic, linearly elastic material.

Problem 4.5

For steel, $E = 30 \cdot 10^6$ and $G = 12 \cdot 10^6$ (force/length²). The components of strain at a point within this material are given by

$$\epsilon = \begin{bmatrix} 0,004 & 0,001 & 0 \\ 0,001 & 0,006 & 0,004 \\ 0 & 0,004 & 0,001 \end{bmatrix}.$$

Compute the corresponding components of the stress tensor.

Lecture 9: Thermal strains

23. Marts 2026

Problem 4.9

Consider an isotropic elastic medium subject to a hydrostatic state of stress:

$$\sigma^{(1)} = \sigma^{(2)} = \sigma^{(3)} = -p.$$

Show that for this state of stress

$$p = -KR_1,$$

where K is the bulk modulus and R_1 is the cubical strain. Consider the implications of $\nu > 1/2$ and $\nu < -1$.

Problem 4.10

For an elastic medium subject to a state of stress σ_{ij} , assume that the deformation is incompressible for $\sigma_{(ii)} \neq 0$.

- (a) Verify that $\nu = 0,5$.
 - (b) Assume also that $\epsilon_{33} = 0$ and determine ν .
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Problem 5.1a

The stress components at a point $P(x, y, z)$ in an isotropic medium are

$$\begin{aligned}\sigma_{xx} &= y + 3z^2, & \sigma_{xy} &= z^3, \\ \sigma_{yy} &= x + 2z, & \sigma_{xz} &= y^2, \\ \sigma_{zz} &= 2x + y, & \sigma_{yz} &= x^2.\end{aligned}$$

No body forces are present.

Does this stress field satisfy equilibrium?

Problem 5.3

For the following stress field,

$$\begin{aligned}\sigma_{xx} &= 80x^3 + y, & \sigma_{xy} &= 1000 + 100y^2, \\ \sigma_{yy} &= 100x^3 + 1600, & \sigma_{xz} &= 0, \\ \sigma_{zz} &= 90y^2 + 100z^3, & \sigma_{yz} &= xz^3 + 100x^2y.\end{aligned}$$

- (a) What is the body force distribution required for equilibrium?
 - (c) With $E = 30 \cdot 10^6$ (force/length²) and $\nu = 0,3$, what is the strain at $(2, 2, 5)$?
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